Consequences of the Entropy-Degree Theorem for Alexandrov Spaces

(arXiv:2606.30403)

An Independent Machine-Checked Formalization Blueprint in Lean 4

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Chapter 1

Analytic Degree, Theorem 1.1, and Core Reduction Lemmas

This chapter establishes the degree-theoretic foundations (§5 of `arXiv:2606.30403`), bridges the main theorem of `arXiv:2606.30403` (Theorem 1.1) and Remark 1.15 to the upstream `RCDEntropyDegree.RCD` library, and isolates the reusable strictness and metric-doubling meta-lemmas in `AlexandrovConsequences/Foundation.lean`.

Definition 1.1 (Analytic Degree and Mitsuishi Comparison Bridge)**.** Let (X, d, μ_X) be a finite-measure space and Y a closed oriented Riemannian N-manifold with volume $\mathrm{Vol}(Y) > 0$. The *analytic degree* of a map $f : X \to Y$ with signed metric Jacobian sJac_f is

$$\deg_{\mathrm{an}}(f) := \frac{1}{\mathrm{Vol}(Y)} \int_X \mathrm{sJac}_f(x)\, d\mu_X(x).$$

A *Mitsuishi degree bridge* packages Y together with a fundamental current, singular homology degree $\deg_{\mathrm{top}}(f)$, current pushforward degree $\deg_{\mathrm{curr}}(f)$, and the naturality identity $\deg_{\mathrm{curr}}(f) = \deg_{\mathrm{top}}(f)$ induced by Mitsuishi's comparison isomorphism $\Psi : H_N^{\mathbf{I}}(X) \xrightarrow{\sim} H_N(X;\mathbb{Z})$.

Lemma 1.2 (Analytic Degree Integral Bound)**.** *For any integrable signed metric Jacobian* sJac_f *and* $\mathrm{Vol}(Y) > 0$, $|\deg_{\mathrm{an}}(f)|\ \mathrm{Vol}(Y) \le \int_X |\mathrm{sJac}_f(x)|\, d\mu_X(x)$.

Proof. Cancel $\mathrm{Vol}(Y)^{-1} \cdot \mathrm{Vol}(Y) = 1$ using $\mathrm{Vol}(Y) > 0$ and apply the triangle inequality for the Bochner integral. □

Theorem 1.3 (Analytic Degree Equals Topological Degree on Class $\mathcal{S}$ Spaces)**.** *On any class* $\mathcal{S}$ *space* X *(compact orientable* N*-metric space with* $\mathcal{H}^{N-1}(\mathcal{S}_X) = 0$ *and Mitsuishi comparison isomorphism* Ψ*), every Lipschitz map* $f : X \to Y$ *satisfies Kirchheim's signed area formula and hence*

$$\deg_{\mathrm{an}}(f) = \deg_{\mathrm{top}}(f).$$

Proof. Substituting Kirchheim's signed area identity $\int_X \mathrm{sJac}_f\, d\mathcal{H}^N = \deg_{\mathrm{curr}}(f)\, \mathcal{H}^N(Y)$ into $\deg_{\mathrm{an}}(f)$ cancels $\mathcal{H}^N(Y)$, and Mitsuishi naturality yields $\deg_{\mathrm{top}}(f)$. □

Theorem 1.4 (Theorem 1.1: Alexandrov Entropy-Degree Theorem and Smoothness Rigidity)**.** *Let* X *be a closed orientable* N*-dimensional Alexandrov space without boundary (*$\mathrm{curv} \ge \kappa$*)*

and $f : X \to Y$ a continuous map into a closed orientable negatively curved locally symmetric N-manifold Y. Via Lipschitz approximation $f_{\mathrm{lip}} \simeq f$,

$$|\deg(f)|\, h(\widetilde{Y})^N\, \mathcal{H}^N(Y) \le h(\overline{X})^N\, \mathcal{H}^N(X).$$

Furthermore, if equality holds together with $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \ge 1$, then the rescaled limit map $\hat{F}_\infty : (X, c\,\mathsf{d}_X) \to \widehat{Y}$ ($c = h(\overline{X})/h(\widetilde{Y}) > 0$) is a surjective global homothety onto the degree-$|\deg(f)|$ cover $\widehat{Y}$, and hence $\mathcal{S}(X) = \emptyset$.

Proof. Homotopy invariance equates $\deg(f) = \deg(f_{\mathrm{lip}})$ in `RCDEntropyDegree.RCD.alexandrov_entropy_degr` and equality rigidity transfers smoothness from $\widehat{Y}$ via `RCDEntropyDegree.RCD.rcd_entropy_degree_equality_` □

Lemma 1.5 (Remark 1.15: Necessity of the Index Condition and Connected-Sum Counterexample)**.** *In the covering factorization $|\deg(f)| = |\deg(\hat{f})| \cdot [\pi_1(Y) : f_*\pi_1(X)]$ with $\deg(f) \ne 0$, we have $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \iff |\deg(\hat{f})| = 1$. For the pinch map $Y \# Y \to Y \vee Y \to Y$ ($b_1(Y) \ge 1$), $|\deg(f)| = 2 > 1 = [\pi_1(Y) : f_*\pi_1(X)]$ and $b_1(Y \# Y) = 2b_1(Y) \ne b_1(Y)$.*

Proof. Cancel the positive integer index $[\pi_1(Y) : f_*\pi_1(X)] \ge 1$ in the degree factorization, and compare $2b_1(Y) \ne b_1(Y)$ for $b_1(Y) \ge 1$. □

Lemma 1.6 (Strictness, Entropy Bounds, and Metric Doubling Cancellation)**.**

1. *Whenever $\mathcal{S}_X \ne \emptyset$, $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \ge 1$, and equality forces $\mathcal{S}_X = \emptyset$, the entropy-volume inequality is strict ($<$).*
2. *Whenever $0 \le h(\overline{X}) \le h_0$ ($h_0 > 0$) and $h(\widetilde{Y}) = h_0$, Theorem 1.1 yields $|\deg(f)|\,\mathrm{Vol}(Y) \le \mathcal{H}^N(X)$ (and $<$ when strict).*
3. *Under Perelman/Petrunin doubling $\mathcal{H}^N(DZ) = 2\mathcal{H}^N(Z)$ and $\mathcal{H}^N(DY) = 2\mathcal{H}^N(Y)$, cancelling $2h_0^N > 0$ gives $|k|\mathcal{H}^N(Y) \le \mathcal{H}^N(Z)$.*

Proof. Any equality contradicts $\mathcal{S}_X \ne \emptyset$; monotonicity of $x \mapsto x^N$ on $[0, \infty)$ and positivity of h_0^N and $2h_0^N$ give the volume and doubling bounds. □

Chapter 2

Branched Coverings and Einstein 4-Orbifold Obstructions

This chapter formalizes Corollaries 1.2 and 1.3 of `arXiv:2606.30403` in `AlexandrovConsequences/BranchedCo`

Definition 2.1 (Cyclic Branched Cover Cone Angle and Density Drop)**.** For a d-fold cyclic branched cover $M_d \to M$ $(d \geq 2)$, the quotient $X = M_d/\mathbb{Z}_d$ carries normal cone angle $\alpha = 2\pi/d \leq \pi < 2\pi$ along the branch locus Σ, giving Bishop–Gromov density $\theta_N(x_0) = \alpha/(2\pi) < 1$ and hence $\mathcal{S}_X \neq \emptyset$.

Corollary 2.2 (Corollary 1.2: Branched Coverings and Hamenstädt's Question)**.** *Let M be a closed hyperbolic N-manifold ($N \geq 3$) and $X = M_d/\mathbb{Z}_d$ ($d \geq 2$) the orbifold quotient of a branched cover along a nonempty codimension-2 locus Σ carrying a negatively curved metric ($h(\overline{X}) \leq N-1$). Then*

$$\mathcal{H}^N(M_d/\mathbb{Z}_d) > \mathrm{Vol}(M) \qquad \textit{and} \qquad \mathrm{Vol}(M_d) > d \cdot \mathrm{Vol}(M).$$

Proof. Since $\alpha = 2\pi/d < 2\pi$ forces $\mathcal{S}_X \neq \emptyset$, the strictness and matched-entropy lemmas give $\mathcal{H}^N(X) > \mathrm{Vol}(M)$, and multiplying by $d \geq 2$ yields $\mathrm{Vol}(M_d) = d\,\mathcal{H}^N(X) > d\ \mathrm{Vol}(M)$. □

Definition 2.3 (Satake Chern–Gauss–Bonnet and Einstein 4-Orbifold Package)**.** With Chern–Gauss–Bonnet constant $C_4 = 4\pi^2/3 > 0$, a 4-orbifold (X, g_E) with $\mathrm{Ric}(g_E) = -3g_E$ satisfies $h(\overline{X}) \leq 3$ and Satake's inequality $\mathrm{Vol}(X, g_E) \leq \frac{4\pi^2}{3}\chi_{\mathrm{orb}}(X)$, while a hyperbolic 4-manifold Y satisfies $h(\widetilde{Y}) = 3$ and $\mathrm{Vol}(Y) = \frac{4\pi^2}{3}\chi(Y)$.

Corollary 2.4 (Corollary 1.3(a)–(b): Einstein 4-Orbifold Euler Obstructions)**.** *Under a continuous map $f : X \to Y$ as above, $\chi_{\mathrm{orb}}(X) \geq |\deg(f)|\,\chi(Y)$. Moreover, if $\mathcal{S}_X \neq \emptyset$ and $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] \geq 1$ (forcing $\deg(f) \neq 0$), then strict inequality holds: $\chi_{\mathrm{orb}}(X) > |\deg(f)|\,\chi(Y)$.*

Proof. Cancel $3^4 = 81$ via matched entropy $h(\overline{X}) \leq 3$ and cancel $4\pi^2/3 > 0$ in the Satake Chern–Gauss–Bonnet inequalities; nonempty $\mathcal{S}_X$ upgrades $\leq$ to $<$. □

Corollary 2.5 (Corollary 1.3(c): Gromov–Thurston Branch-Locus Euler Sign)**.** *For $X = M_d/\mathbb{Z}_d$ ($d \geq 2$) branched along a closed surface $\Sigma \subset M$, any Einstein metric $\mathrm{Ric}(g_E) = -3g_E$ on X forces $\chi(\Sigma) < 0$.*

Proof. Combining the orbifold Riemann–Hurwitz identity $\chi_{\mathrm{orb}}(X) = \chi(M) - (1-1/d)\chi(\Sigma)$ with $\chi_{\mathrm{orb}}(X) > \chi(M)$ and $1 - 1/d > 0$ yields $\chi(\Sigma) < 0$. □

Chapter 3

Cone-Manifolds, Convex Cores, Weil–Petersson Bounds, and Mapping Tori

This chapter formalizes Corollaries 1.4–1.8 of `arXiv:2606.30403` in `AlexandrovConsequences/ConvexCoresAn`

Corollary 3.1 (Corollary 1.4: Cone-Manifolds with Cone Angles $\leq 2\pi$)**.** *For a compact orientable N-cone-manifold Z ($N \geq 3$) with cone angles $\leq 2\pi$ and $K_{\mathrm{reg}} \geq -1$ (so $0 \leq h(\overline{Z}) \leq N-1$), every continuous map $f : Z \to Y$ satisfies $|\deg(f)|\, h(\widetilde{Y})^N \mathcal{H}^N(Y) \leq (N-1)^N \mathcal{H}^N(Z)$.*

Proof. Direct application of `upper_entropy_bound_volume_inequality` with $h_{\mathrm{upper}} = N - 1$. □

Corollary 3.2 (Corollary 1.5: Hyperbolic Manifolds with Locally Convex Boundary)**.** *For a compact oriented N-manifold Z ($N \geq 3$) with locally convex boundary and $\sec \geq -1$ mapped with relative degree $k \in \mathbb{Z}$ to a hyperbolic N-manifold Y_{geod} with totally geodesic boundary, Perelman doubling $DZ \to DY_{\mathrm{geod}}$ yields $|k|\ \mathrm{Vol}(Y_{\mathrm{geod}}) \leq \mathrm{Vol}(Z)$.*

Proof. Since $N \geq 3$, $h_0 = N - 1 > 0$; apply `metric_doubling_volume_degree_bound`. □

Corollary 3.3 (Corollary 1.6: Geometrically Finite Hyperbolic 3-Manifold Convex Cores)**.** *For convex cores $C_M \to C_{M_g}$ of geometrically finite hyperbolic 3-manifolds ($N = 3, h_0 = 2$) of relative degree k, both the doubled core and smooth r-neighborhood approximation limit yield $|k|\ \mathrm{Vol}(C_{M_g}) \leq \mathrm{Vol}(C_M)$.*

Proof. Specialize Corollary 1.5 to $N = 3$ and pass to the limit $\mathrm{Vol}(Z_{r_i}) \to \mathrm{Vol}(C_M)$ via `ge_of_tendsto`. □

Corollary 3.4 (Corollary 1.7: Unified Degree and Weil–Petersson Core Volume Bound)**.** *Combining Corollary 1.6 for $f \circ h : M'_{\mathrm{geod}} \to M_{\mathrm{geod}}$ ($|\deg(h)| = 1$) with the Bridgeman–Brock–Bromberg comparison gives*

$$|k|\ \mathrm{Vol}(C_{M_{\mathrm{geod}}}) + A(\partial Z')\, d_{\mathrm{WP}}(\partial_c M', \partial_c M'_{\mathrm{geod}}) - \delta \leq \mathrm{Vol}(C_{M'}).$$

Proof. Degree multiplicativity gives $|\deg(f \circ h)| = |k|$, and adding the Weil–Petersson comparison $\mathrm{Vol}(C_{M'}) - \mathrm{Vol}(C_{M'_{\mathrm{geod}}}) \geq A(\partial Z')\, d_{\mathrm{WP}} - \delta$ completes the proof. □

Definition 3.5 (Infimal Convex-Hyperbolic Volume)**.** For a parameterized family S of convex hyperbolic metrics on the compactified end-periodic mapping torus $\overline{M}_\phi$, define $\underline{\mathrm{Vol}}(\overline{M}_\phi) := \inf_{g \in S} \mathrm{Vol}(\overline{M}_\phi, g)$.

Corollary 3.6 (Corollary 1.8: Mapping-Class Translation Length Lower Bound)**.** *On a nonempty family S of convex hyperbolic metrics on $\overline{M}_\phi$, Corollary 1.5 holds uniformly, giving $|k|\ \mathrm{Vol}(Y) \le \underline{\mathrm{Vol}}(\overline{M}_\phi)$, and combining with $\underline{\mathrm{Vol}}(\overline{M}_\phi) \le V_{\mathrm{oct}}\,\tau(\phi)$ ($V_{\mathrm{oct}} > 0$) yields*

$$\frac{|k|}{V_{\mathrm{oct}}}\ \mathrm{Vol}(Y) \le \tau(\phi).$$

Proof. Apply `le_csInf` over the nonempty image `vol '' S`, then divide by $V_{\mathrm{oct}} > 0$. □

Chapter 4

Simplicial Volume and Boundary Volume Rigidity

This chapter formalizes Theorems 1.9 and 1.10 of `arXiv:2606.30403` in `AlexandrovConsequences/Simplicial`

Definition 4.1 (Gromov's Simplicial Volume)**.** Given fundamental-cycle representatives `Cycles` with ℓ^1-norm functional `l1Norm`, $\|DM\| := \inf_{c \in \mathrm{Cycles}} \|c\|_1 \geq 0$.

Theorem 4.2 (Theorem 1.9: Acylindrical Convex-Core Bound via Storm's Theorem)**.** *For a relative degree-k map $f : (Z, \partial Z) \to (M, \partial M)$ into a compact acylindrical 3-manifold M with Thurston geodesic-boundary model $\psi : M \to M_{\mathrm{geod}}$ ($|\deg(\psi)| = 1$), Perelman doubling of C_Z and Storm's identity $\mathrm{Vol}(DC_{M_{\mathrm{geod}}}) = \|DM\|$ (or $v_3\|DM\|$) yield $\mathrm{Vol}(C_Z) \geq \frac{|k|}{2}\|DM\|$.*

Proof. Use $|\deg(\psi \circ f)| = |k|$ and matched entropy $h_0 = 2$ on the doubles, then divide by 2. □

Theorem 4.3 (Theorem 1.10: Boundary Volume Inequality and Rigidity)**.** *For a compact orientable N-dimensional ($N \geq 3$) $\mathrm{Alex}(-1)$ space X with boundary mapped with degree $k \neq 0$ to a compact hyperbolic N-manifold Y with totally geodesic boundary, $\mathrm{Vol}(X) \geq |k|\,\mathrm{Vol}(Y)$. Moreover, if $\mathrm{Vol}(X) = |k|\,\mathrm{Vol}(Y)$ and $|k| = [\pi_1(DY) : (Df)_*\pi_1(DX)]$, then $h(\overline{DX}) = N - 1$ (homothety factor $c = 1$), $\mathcal{S}_{DX} = \emptyset$, DX is globally isometric to the degree-$|k|$ cover $\widehat{DY}$, $\partial X = \mathrm{Fix}(\iota_{DX})$ is totally geodesic, and $|k| = 1$ implies $X \cong Y$.*

Proof. Volume equality $\mathrm{Vol}(DX) = |k|\,\mathrm{Vol}(DY) > 0$ together with Theorem 1.1 sandwiches $(N-1)^N \leq h(\overline{DX})^N \leq (N-1)^N$, forcing $h(\overline{DX}) = N - 1$ by strict monotonicity of $x \mapsto x^N$ on $[0, \infty)$ and triggering unrescaled ($c = 1$) rigidity. □

Chapter 5

Gromov–Hausdorff Volume and Entropy-Volume Stability

This chapter formalizes Corollaries 1.11 and 1.12 of `arXiv:2606.30403` in `AlexandrovConsequences/GromovHa`

Lemma 5.1 (Reviron Entropy Lower-Semicontinuity Product Limit)**.** *If* $\mathrm{Vol}(M_i) \to V_X > 0$, $h(\widetilde{M}_i)^N \mathrm{Vol}(M_i) \to L$, *and* $\forall \varepsilon > 0, \forall^\infty i$, $h_{\lim} \le h(\widetilde{M}_i) + \varepsilon$, *then* $h_{\lim}^N V_X \le L$.

Proof. Since $V_X > 0$, $h(\widetilde{M}_i)^N \to L/V_X$. For each $\varepsilon > 0$, eventually $(\max(h_{\lim} - \varepsilon, 0))^N \le h(\widetilde{M}_i)^N$, so taking $i \to \infty$ and then $\varepsilon \to 0^+$ yields $h_{\lim}^N \le L/V_X$. □

Corollary 5.2 (Corollary 1.11: Gromov–Hausdorff Stability of Hyperbolic Volume)**.** *If closed* N*-manifolds* (M_i) *(*$N \ge 3$, $\sec \ge -1$*) converge in Gromov–Hausdorff distance to* (X, d) *with* $\mathrm{Vol}(M_i) \to k\ \mathrm{Vol}(Y)$ *(*$k \ge 1$*) and Perelman stability transfers* $|\deg| = \mathrm{index} = k$, *then* $\mathrm{Vol}(X) = k\ \mathrm{Vol}(Y)$, $h(\widetilde{X}) = N - 1$, $\mathcal{S}_X = \emptyset$, *and* X *is globally isometric to the smooth degree-*k *hyperbolic cover of* Y *(*Y *itself when* $k = 1$*).*

Proof. Uniqueness of Hausdorff limits equates $\mathrm{Vol}(X) = k\ \mathrm{Vol}(Y)$; Perelman stability transfers degree/index k; and Theorem 1.1 squeezes $h(\widetilde{X}) = N - 1$, triggering unrescaled rigidity. □

Corollary 5.3 (Corollary 1.12: Locally Symmetric GH Entropy-Volume Stability)**.** *For any closed orientable negatively curved locally symmetric* N*-manifold* Y *(*$N \ge 3$*), if* $h(\widetilde{M}_i)^N \mathrm{Vol}(M_i) \to k\, h(\widetilde{Y})^N \mathrm{Vol}(Y)$ *(*$k \ge 1$*),* $\mathrm{Vol}(M_i) \to \mathrm{Vol}(X) > 0$, *and Reviron's entropy lower-semicontinuity holds, then* $h(\widetilde{X})^N \mathrm{Vol}(X) = k\, h(\widetilde{Y})^N \mathrm{Vol}(Y)$, $h(\widetilde{X}) > 0$, $\mathcal{S}_X = \emptyset$, *and* $(X, (h(\widetilde{X})/h(\widetilde{Y}))\mathsf{d}_X)$ *is globally isometric to the smooth degree-*k *cover* $\widehat{Y}$.

Proof. Combine Lemma 5.1 ($\le$) with Theorem 1.1 ($\ge$) to obtain equality, extract $h(\widetilde{X}) > 0$, and apply Theorem 1.1 equality rigidity. □

Chapter 6

Cone-Manifold and Orbifold Global Volume Minimality

This chapter formalizes Corollaries 1.13–1.15 (`ConeAndOrbifoldMinimality.lean`) and the unconditional octonionic and cone/orbifold master certificates (`MasterCatalog.lean`).

Corollary 6.1 (Corollary 1.13: Simplicial Volume of Alexandrov Spaces and Cone-Manifolds)**.** *Using Gromov–Thurston proportionality* $\mathrm{Vol}(Y) = v_N\|Y\|$*, we have* $\mathcal{H}^N(X) \geq |\deg(f)|v_N\|Y\|$*. For a homotopy equivalence,* $\mathcal{H}^N(X) \geq v_N\|X\|$ *with equality iff* $X \cong Y$*, and* $\mathcal{S}_X \neq \emptyset$ *forces strict inequality* $\mathcal{H}^N(X) > v_N\|X\|$*.*

Proof. Substitute $\mathrm{Vol}(Y) = v_N\|Y\|$ into the matched-entropy ($h_0 = N - 1$) and strictness lemmas. □

Corollary 6.2 (Corollary 1.14: Global Cone-Metric Volume Minimality on Hyperbolic Manifolds)**.** *On any closed hyperbolic* N*-manifold* (M, g_{hyp}) *(*$N \geq 3$*), every hyperbolic cone-metric* g_{cone} *with cone angles* $\leq 2\pi$ *and nonempty singular locus satisfies* $\mathcal{H}^N(M, g_{\mathrm{cone}}) > \mathrm{Vol}(M, g_{\mathrm{hyp}})$*.*

Proof. Apply Corollary 1.13 to $\mathrm{id}_M : (M, g_{\mathrm{cone}}) \to (M, g_{\mathrm{hyp}})$. □

Lemma 6.3 (Selberg Finite Cover Cone-Angle Unwrapping)**.** *Along a torsion-free Selberg finite cover* $p : M \to \mathcal{O}$ *with local isotropy orders* $m_i \geq 1$*, the lifted cone angle satisfies* $m_i\alpha_i \leq 2\pi \iff \alpha_i \leq 2\pi/m_i$ *(and similarly for* $<$*).*

Proof. Multiply or divide both sides by $(m_i : \mathbb{R}) > 0$. □

Corollary 6.4 (Corollary 1.15: Orbifold Cone-Metric Global Volume Minimality)**.** *On a compact orientable negatively curved locally symmetric* N*-orbifold* $(\mathcal{O}, g_{\mathrm{sym}})$ *(*$N \geq 3$*), any singular cone-metric* g_{cone} *with* $K_{\mathrm{reg}} \geq -\kappa_0^2$ *and cone angles* $\alpha_i \leq 2\pi/m_i$ *satisfies* $\mathcal{H}^N(|\mathcal{O}|, g_{\mathrm{cone}}) \geq \mathrm{Vol}(\mathcal{O}, g_{\mathrm{sym}})$*, with strict inequality whenever some* $\alpha_i < 2\pi/m_i$*.*

Proof. Lift to a degree-d ($d \geq 1$) Selberg manifold cover M, apply Theorem 1.1 (or its strictness corollary when $m_i\alpha_i < 2\pi$) to id_M, and divide by $d > 0$. □

Theorem 6.5 (Unconditional Octonionic ($\mathbb{O}\mathbf{H}^2$) and Master Cone/Orbifold Synthesis)**.** *Composing* `BCG.octonionic_bcg_main_theorem` *($N = 16, h_0 = 22$ on $\mathbb{O}\mathbf{H}^2$ via Ruan's 2024 repair) with Corollary 1.12 gives unconditional octonionic entropy-degree rigidity and Gromov–Hausdorff stability; bundling Corollaries 1.2, 1.14, and 1.15 gives a unified global volume minimality certificate.*

Proof. Apply `BCG.octonionic_bcg_main_theorem` to discharge the rank-one inequality at $(N, h_0) = (16, 22)$, feed it into `gh_locally_symmetric_entropy_volume_stability`, and assemble the three strict volume minimality theorems. □